8.0 Pre-learning: Angles and Parallel Lines

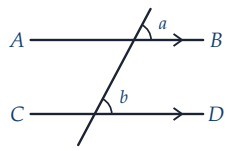
預備知識 · 平行線與角

Every result about quadrilaterals in this chapter rests on three facts about parallel lines. Learn to spot the F, the Z and the U, and the rest of the chapter becomes easy.

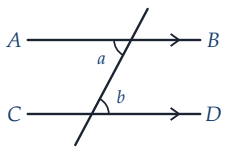
A Three angle pairs 截線造成的三對角

Concept 概念

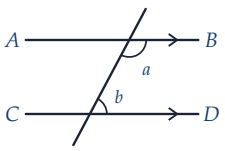
A **transversal**（截線） is a straight line that cuts two other lines. When the two lines are **parallel**（平行） (marked with arrows ›), the transversal creates three special pairs of angles.



Corresponding angles
同位角 · F-shape · $a = b$



Alternate angles
錯角 · Z-shape · $a = b$



Interior angles
同旁內角 · U-shape · $a + b = 180°$

即係話

一條直線「斬」過兩條平行線，就會整出好多個角。我哋唔使逐個記，只要認得三個字母形狀：**F**、**Z**、**U**。

點樣認？

用筆沿住兩個角嘅邊描一次：描到似 F 就係同位角；似 Z 就係錯角；似 U（兩個角喺兩條平行線中間、同一邊）就係同旁內角。

Key points 重點記住

1. **F** and **Z**: the two angles are **equal**.
2. **U**: the two angles are **supplementary**（互補）, i.e. they add up to $180°$.
3. These facts only work when the two lines are **parallel**. Always look for the arrow marks first.

口訣

「F 同 Z，一樣大；U 字形，加埋一百八。」

B Two directions: using vs. proving 性質 vs 判定：兩個方向

The same angle facts work in **two directions**. Your reason must show which direction you are going.

Angle pair	Know $AB \parallel CD$ → find angles	Know angles → prove $AB \parallel CD$
Corresponding (F)	$a = b$ (corr. ∠s, AB ∥ CD)	$a = b$ (corr. ∠s equal)
Alternate (Z)	$a = b$ (alt. ∠s, AB ∥ CD)	$a = b$ (alt. ∠s equal)
Interior, same side (U)	$a + b = 180°$ (int. ∠s, AB ∥ CD)	$a + b = 180°$ (int. ∠s supp.)

即係話

左邊：已知平行，用嚟**計角**，理由要寫埋「AB ∥ CD」。
右邊：已知角，用嚟**證平行**，理由寫「equal」或者「supp.」。

生活例子

好似「落雨 → 地濕」同「地濕 → 證明落過雨？」，方向唔同，講法就唔同。

C Traps to avoid 陰濕位

✗ To prove $AB \parallel CD$: $\angle a = \angle b$ (alt. ∠s, AB ∥ CD)	✓ To prove $AB \parallel CD$: $\angle a = \angle b$, ∴ AB ∥ CD (alt. ∠s equal)
✗ The lines look parallel, so $x = 70°$ (corr. ∠s).	✓ Check for arrow marks or the words "$AB \parallel CD$" first. No ∥, no angle fact.
✗ Interior angles are equal: $x = 110°$.	✓ Interior angles **add up to** $180°$: $x = 180° - 110° = 70°$.

點解錯？

第一個：要證平行，你就唔可以用「AB ∥ CD」做理由，咁即係用個答案證個答案，循環論證，扣分。

第二個：「睇落平行」唔等於平行。冇箭咀、題目冇講，就唔准用。

第三個：U 形係加埋 180°，唔係相等，好多人中伏。

D Worked example 例題

Example Level 1 AB and CD are cut by the straight line EF at G and H. Given $\angle EGB = 112°$ and $\angle GHC = 68°$, prove that $AB \parallel CD$.

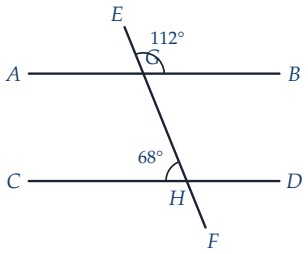
$\angle BGH = 180° - \angle EGB$ *(adj. ∠s on st. line)*

$= 180° - 112°$

$= 68°$

$\therefore \angle BGH = \angle GHC = 68°$

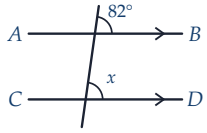
$\therefore$ **AB ∥ CD** *(alt. ∠s equal)*



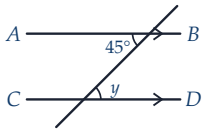
思路

112° 同 68° 唔係一對 F/Z/U。先用「直線上嘅鄰角」計出 ∠BGH，佢同 ∠GHC 砌成 Z 形，就證到平行。

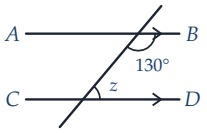
E Basic drilling 基本練習



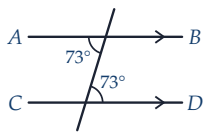
1 $AB \parallel CD$. Find x.



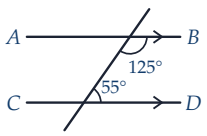
2 $AB \parallel CD$. Find y.



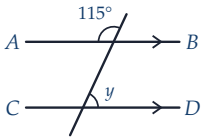
3 $AB \parallel CD$. Find z.



4 Prove that $AB \parallel CD$.



5 Prove that $AB \parallel CD$.



6 $AB \parallel CD$. Find y. (two steps)

Answers （答案） **1** $x = 82°$ (corr. ∠s, AB ∥ CD) · **2** $y = 45°$ (alt. ∠s, AB ∥ CD) · **3** $z = 50°$ (int. ∠s, AB ∥ CD) · **4** alt. ∠s equal, so AB ∥ CD · **5** $125° + 55° = 180°$, so AB ∥ CD (int. ∠s supp.) · **6** $\angle AXY = 65°$ (adj. ∠s on st. line), $y = 65°$ (alt. ∠s, AB ∥ CD)

8.1 Basic Knowledge of Quadrilaterals

四邊形的基本知識

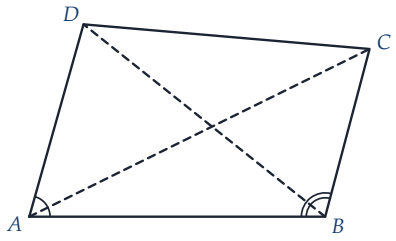
Before studying special quadrilaterals, we need a common language: which sides and angles are "neighbours", which are "opposite", and what a diagonal is.

A Parts of a quadrilateral 四邊形的組成部分

Concept 概念

A **quadrilateral**（四邊形） is a polygon with four sides. We name it by going round its vertices **in order**, e.g. *ABCD*.

Term	Meaning	In *ABCD*
adjacent sides（鄰邊）	share a common vertex	*AB*, *BC*
opposite sides（對邊）	not adjacent	*AB*, *DC*
adjacent angles（鄰角）	share a common side	$\angle A$, $\angle B$
opposite angles（對角）	not adjacent	$\angle A$, $\angle C$
diagonal（對角線）	joins two non-adjacent vertices	*AC*, *BD*

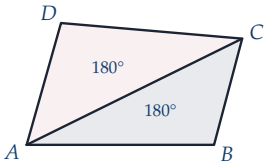


生活化理解

當四個頂點係四間屋：**隔籬屋**就係「鄰」；**對面屋**就係「對」；由一間屋**斜斜行過對面**嗰條路，就係對角線。

Key points 重點記住

1. Every quadrilateral has **2 diagonals**, **2 pairs** of opposite sides and **2 pairs** of opposite angles.
2. **Angle sum of a quadrilateral**（四邊形內角和） $= (4-2)\times 180° = 360°$ ($\angle$ sum of polygon).
3. Why 360°? One diagonal cuts it into **two triangles**, and each triangle gives $180°$.



✗ Quadrilateral *ACBD* (letters out of order).

✓ Name the vertices in order round the figure: *ABCD*.

✗ *AB* is a diagonal of *ABCD*.

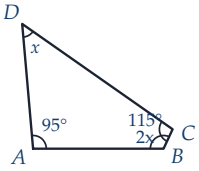
✓ *AB* is a **side**. The diagonals are *AC* and *BD*.

口訣

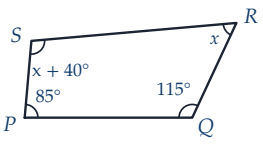
「同一點就係鄰，唔鄰就係對，斜斜過對面就係對角線。」

B Basic drilling 基本練習

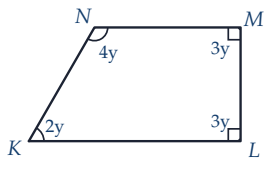
1 In *PQRS*, name the side opposite to *PQ*.



2 In *PQRS*, name both diagonals.



3 Can a quadrilateral have angles 100°, 100°, 100°, 70°?



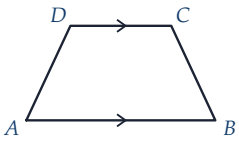
4 Find x.

5 Find x.

6 Find y.

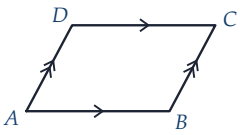
Answers（答案） **1** *SR* · **2** *PR* and *QS* · **3** No: sum = 370° ≠ 360° · **4** $95° + 115° + 2x + x = 360°$, $x = 50°$ · **5** $85° + 115° + x + x + 40° = 360°$, $x = 60°$ · **6** $2y + 3y + 3y + 4y = 360°$, $y = 30°$

C Five special quadrilaterals 五種特殊四邊形



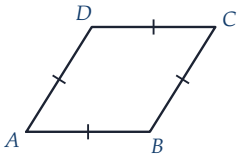
Trapezium（梯形）: a quadrilateral with **only one** pair of parallel opposite sides.

即係話：得**一對**對邊平行（唔係兩對）。



Parallelogram（平行四邊形）: two pairs of parallel opposite sides.

即係話：**兩對**對邊都平行，好似一個被人推歪咗嘅長方形。



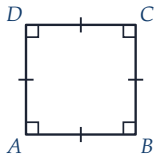
Rhombus（菱形）: four equal sides.

即係話：**四條邊一樣長**，好似一粒「菱角」或者撲克牌嘅階磚。



Rectangle（長方形）: four equal interior angles (each $90°$).

即係話：**四隻角一樣大**，所以每隻都係 $360° \div 4 = 90°$。



Square（正方形）: four equal sides **and** four equal interior angles.

即係話：菱形同長方形嘅條件**一齊有**。

家族關係

正方形係「全能」：佢**係**菱形，**亦係**長方形，**亦係**平行四邊形。

但倒轉講就唔得：菱形**唔一定**係正方形。

記法

睇定義只問兩件事：

① 邊：幾多對平行？幾條一樣長？

② 角：四隻角係咪一樣大？

小心

本課程嘅梯形係「**只有**一對」平行邊，所以平行四邊形**唔係**梯形。

標記點睇

箭咀 › = 平行（兩組就用 › 同 »）。

短劃 | = 一樣長。

小正方形 ∟ = 直角 90°。

睇圖第一步：先搵晒呢三種記號。

Traps 陰濕位

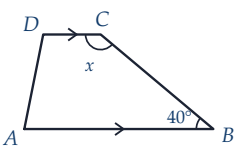
✗ A parallelogram is a special trapezium.	✓ A trapezium has **only one** pair of parallel sides, so a parallelogram is **not** a trapezium.
✗ A rhombus is a square.	✓ A **square** is always a rhombus, but a rhombus need not have right angles.
✗ A rectangle has four equal sides.	✓ A rectangle has four equal **angles**. Its sides need not all be equal.

點解錯？

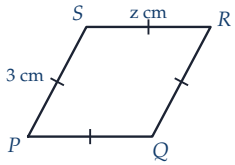
定義講「**邊**」定「**角**」要睇清楚：菱形講邊，長方形講角，正方形兩樣都要。

考試問「Is ABCD a square?」，要兩樣條件都證到先得。

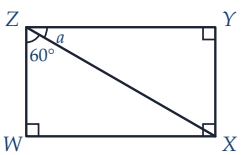
D Basic drilling: use the definitions 基本練習：用定義求未知數



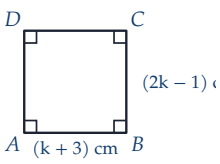
1 Trapezium, $DC \parallel AB$. Find x.



2 Rhombus $PQRS$. Find z.



3 Rectangle $WXYZ$. Find a.



4 Square $ABCD$. Find k.

Answers（答案） **1** $x + 40° = 180°$ (int. ∠s, $DC \parallel AB$), $x = 140°$ · **2** $RS = PS$ (by definition), $z = 3$ · **3** $a + 60° = 90°$ (by definition), $a = 30°$ · **4** $k + 3 = 2k - 1$ (by definition), $k = 4$

8.2 Parallelograms

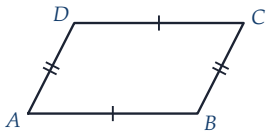
平行四邊形

A parallelogram is defined by its parallel sides, yet it hides three more facts about its sides, angles and diagonals. These three facts are the tools you will use in almost every question of this chapter.

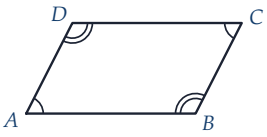
A Properties of parallelograms 平行四邊形的性質

Concept 概念

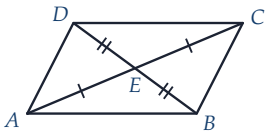
A **parallelogram**（平行四邊形） is a quadrilateral with two pairs of parallel opposite sides. Every parallelogram also has the following three **properties**（性質）.



(a) **Opposite sides are equal**
$AB = DC, AD = BC$



(b) **Opposite angles are equal**
$\angle A = \angle C, \angle B = \angle D$



(c) **Diagonals bisect each other**
$AE = EC, BE = ED$

即係話

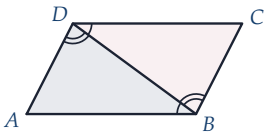
平行四邊形「對住嗰對」一定一樣：對邊一樣長，對角一樣大。兩條對角線喺中間交叉，會互相**平分**（bisect），即係交點係兩條線嘅中點。

小心

「互相平分」唔係「一樣長」！兩條對角線通常**唔等長**，只係各自被切開一半。

Why is it true? 點解會咁？

Join the diagonal BD. By alternate angles ($AB \parallel DC$ and $AD \parallel BC$), the two triangles share two pairs of equal angles and the common side BD, so $\triangle ABD \cong \triangle CDB$ (ASA). Matching parts of congruent triangles give the equal sides and equal angles.



生活化理解

用剪刀沿對角線剪開，兩塊三角形可以**完全疊埋**。疊得埋，對應嘅邊同角自然一樣。

Reasons to write 理由寫法

Property used	Reason code in your solution
opposite sides equal	(opp. sides of ∥gram)
opposite angles equal	(opp. ∠s of ∥gram)
diagonals bisect each other	(diagonals of ∥gram)
adjacent angles add up to $180°$	(int. ∠s, AB ∥ DC)

Key points 重點記住

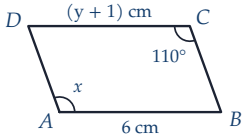
1. Three properties: **opposite sides**, **opposite angles**, **diagonals bisect**.
2. Two **adjacent** angles of a parallelogram add up to $180°$ (they are interior angles).
3. You may use these properties **only after** you know the figure is a parallelogram.

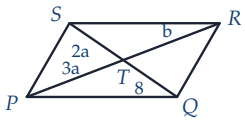
口訣

「對邊等、對角等、對角線互相平分；鄰角加埋一百八。」

B Worked examples: using the properties 例題：運用性質

Example 1 Level 1 Find the unknowns in each parallelogram.





$x = 110°$ (*opp. ∠s of ∥gram*)

$DC = AB$ (*opp. sides of ∥gram*)

$y + 1 = 6$

$\boxed{y = 5}$

$ST = TQ$ (*diagonals of ∥gram*)

$2a = 8$

$\boxed{a = 4}$

$RT = TP$ (*diagonals of ∥gram*)

$b = 3a = \boxed{12}$

思路

(a) 先搵「對住」嘅嘢：∠A 對 ∠C、AB 對 DC。

(b) 有對角線就諗「一半一半」：S 去 T 等於 T 去 Q。

寫法

每一行都要有理由，括號入面寫 reason code。

Example 2 Level 1 $ABCD$ is a parallelogram. E is a point on AB such that $AD = AE$. If $\angle CDE = 58°$, find a and c.

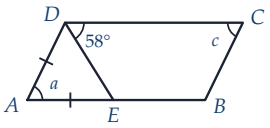
$\angle AED = \angle CDE = 58°$ (*alt. ∠s, AB ∥ DC*)

$\angle ADE = \angle AED = 58°$ (*base ∠s, isos. △*)

$a + 58° + 58° = 180°$ (*∠ sum of △*)

$\boxed{a = 64°}$

$c = a = \boxed{64°}$ (*opp. ∠s of ∥gram*)



思路

58° 喺 D，a 喺 A：中間要搵條橋。AB ∥ DC 就有 Z 形，再用等腰三角形底角相等。

Traps 陰濕位

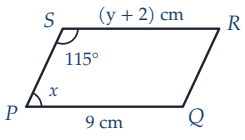
✗ Diagonals of a parallelogram are equal: $AC = BD$.

✓ Diagonals **bisect each other**: $AE = EC$ and $BE = ED$. They are usually **not** equal.

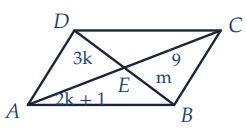
✗ Adjacent angles are equal: $\angle A = \angle B$.

✓ **Opposite** angles are equal. Adjacent angles add up to $180°$.

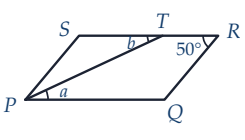
C Basic drilling 基本練習



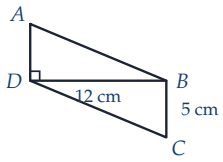
1 ∥gram $PQRS$. Find x and y.



2 ∥gram $ABCD$. Find k and m.



3 ∥gram $PQRS$, PT bisects $\angle SPQ$. Find a and b.

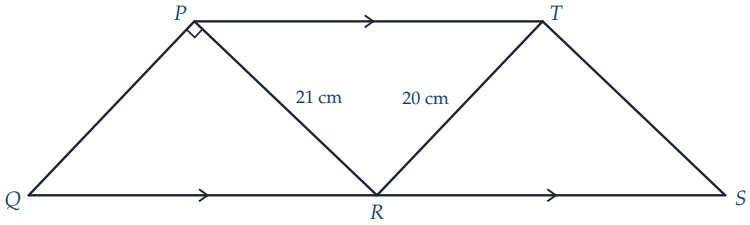


4 ∥gram $ABCD$, $AD \perp DB$. Find the perimeter.

Answers （答案） **1** $x = 65°$ (int. ∠s, PQ ∥ SR), $y = 7$ (opp. sides of ∥gram) · **2** $2k + 1 = 9$, $k = 4$; $m = 3k = 12$ (diagonals of ∥gram) · **3** $\angle SPQ = 50°$ (opp. ∠s), $a = 25°$; $b = a = 25°$ (alt. ∠s, PQ ∥ SR) · **4** $AD = 5$, $AB = \sqrt{5^2 + 12^2} = 13$, perimeter $= 36$ cm

D Worked examples: combining with Pythagoras 例題：配合畢氏定理

Example 3 Level 2 $PT \parallel QS$. R lies on QS such that $RP \parallel ST$, $RT \parallel QP$ and $\angle QPR = 90°$. $TR = 20$ cm and $RP = 21$ cm. Find (a) PT, (b) QS, (c) the perimeter of $PQST$.



(a) $\angle PRT = \angle QPR = 90°$ (*alt. ∠s, RT ∥ QP*)

$PT = \sqrt{20^2 + 21^2}$ cm = **29 cm** (*Pyth. theorem*)

(b) $PQRT$ is a ∥gram, so $QR = PT$ (*by definition; opp. sides of ∥gram*)

$PRST$ is a ∥gram, so $RS = PT$ (*by definition; opp. sides of ∥gram*)

$QS = 29 + 29 =$ **58 cm**

(c) $PQ = TR = 20$ cm, $ST = RP = 21$ cm (*opp. sides of ∥gram*)

Perimeter $= 29 + 20 + 58 + 21 =$ **128 cm**

思路

一個圖入面其實有**兩個**平行四邊形：PQRT 同 PRST。用兩隻顏色筆分別描出嚟，對邊相等就一目了然。

點樣搵？

見到「∥」嘅條件有兩對，就試吓砌出「兩對對邊都平行」嘅四邊形。PT ∥ QR 加 RT ∥ QP → PQRT；PT ∥ RS 加 RP ∥ ST → PRST。

小心

(b) 要先講「PQRT is a ∥gram (by definition)」，之後先可以用 opp. sides of ∥gram。

Example 4 Level 2 $ABCD$ is a parallelogram. AC and BD meet at O. $AB = 5$ cm, $AD = 13$ cm and $OD = 6$ cm. (a) Find $\angle ABD$. (b) Find AC, correct to 3 significant figures.

(a) $OB = OD = 6$ cm, so $BD = 12$ cm (*diagonals of ∥gram*)

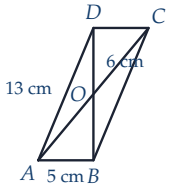
$AB^2 + BD^2 = 25 + 144 = 169 = AD^2$

$\angle ABD = 90°$ (*converse of Pyth. theorem*)

(b) $OA = \sqrt{5^2 + 6^2}$ cm $= \sqrt{61}$ cm (*Pyth. theorem*)

$OC = OA = \sqrt{61}$ cm (*diagonals of ∥gram*)

$AC = 2\sqrt{61}$ cm = **15.6 cm** (3 sig. fig.)



小心

5、12、13 係一組畢氏三元數。見到就要諗：「係咪有直角？」用 **converse of Pyth. theorem** 證，唔係 Pyth. theorem。

Key points 重點記住

1. A complicated figure is often **two or more** parallelograms glued together. Find each one first.
2. State "... is a parallelogram (by definition)" **before** using any property of it.
3. A right angle inside a parallelogram is your cue for **Pythagoras' theorem**.

口訣

「先認∥gram，後用性質；見直角，諗畢氏。」

Traps 陰濕位

✗ $OA \approx 7.81$ cm, so $AC \approx 2 \times 7.81$ cm (rounded in the middle).

✓ Keep $OA = \sqrt{61}$ cm exactly. Round **only** the final answer: $AC = 2\sqrt{61} = 15.6$ cm.

✗ "$5^2 + 12^2 = 13^2$, so $\angle ABD = 90°$ (Pyth. theorem)".

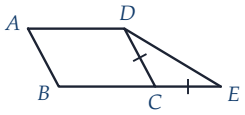
✓ Using side lengths to prove a right angle is the **converse**: (converse of Pyth. theorem).

點解錯？

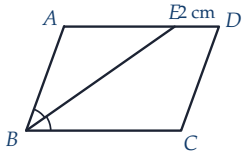
中途四捨五入會累積誤差，好多題會因此差 0.1 失分。

「知直角求邊」用 Pyth. theorem；「知三邊證直角」用 converse。

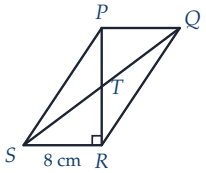
E Basic drilling 基本練習



1 ∥gram $ABCD$, BCE is a straight line, $CD = CE$, $BE = 18$ cm. Find the perimeter of $ABCD$.



2 ∥gram $ABCD$, BE bisects $\angle ABC$, $ED = 2$ cm, perimeter 24 cm. Find AB.



3 ∥gram $PQRS$, $PR \perp RS$, $QS = 20$ cm. Find PR and QR (3 sig. fig.).

4 In ∥gram $ABCD$, $\angle A = 3\angle B$. Find $\angle C$.

5 The diagonals of ∥gram $PQRS$ meet at O. $PR = 14$ cm, $QS = 10$ cm, $PQ = 8$ cm. Find the perimeter of $\triangle OPQ$.

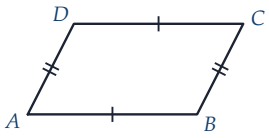
6 True or false: "If the diagonals of $ABCD$ are equal, then $ABCD$ is a parallelogram."

Answers （答案） **1** $AB + BC = CD + BC = CE + BC = 18$ cm; perimeter = 36 cm · **2** ∠AEB = ∠EBC = ∠ABE, so AE = AB; $2(AB + AB + 2) = 24$, $AB = 5$ cm · **3** $TS = 10$, $TR = 6$, $PR = 12$ cm; $QR = PS = \sqrt{208} = 14.4$ cm · **4** $\angle A + \angle B = 180°$ (int. ∠s), $4\angle B = 180°$, $\angle B = 45°$, $\angle C = \angle A = 135°$ · **5** $OP = 7$, $OQ = 5$ (diagonals of ∥gram), perimeter = 20 cm · **6** False (e.g. an isosceles trapezium has equal diagonals)

F Conditions for parallelograms 平行四邊形的判定條件

Concept 概念

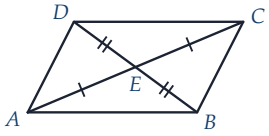
Section A answered "**If** it is a parallelogram, **what** do we know?". Now we turn it round: "**What** must we show **to prove** that a quadrilateral is a parallelogram?". Besides the definition (two pairs of parallel sides), any **one** of the following four conditions is enough.



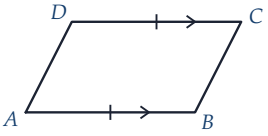
(a) Both pairs of opposite sides equal
(opp. sides equal)



(b) Both pairs of opposite angles equal
(opp. ∠s equal)



(c) Diagonals bisect each other
(diags. bisect each other)



(d) One pair of sides equal **and** parallel
(2 sides equal and ∥)

即係話

上一部分係「已知係∥gram → 用性質」。
今次係**倒轉**：「想證佢係∥gram → 要搵咩證據」。

記法

(a)(b)(c) 其實就係三個性質倒轉講。
(d) 係新嘅：**同一對**邊，要**又等長、又平行**，兩樣都要。

生活化理解

好似認朋友：佢著咩衫係「性質」；你要搵出「證據」先可以話「係佢」。四個條件，有一個成立就足夠。

Key points 重點記住

1. **Properties** (Section A) are used **after** we know it is a parallelogram; **conditions** (this section) are used **to prove** it is one.
2. Condition (d) needs the **same pair** of sides to be both equal **and** parallel.
3. Write the condition as the reason on the **last line**, e.g. "$\therefore$ $ABCD$ is a parallelogram (opp. ∠s equal)".

口訣

「邊、角、對角線，倒轉講就係判定；一對邊，又等又平行。」

Traps 陰濕位

✗ $AB = DC$ and $AD \parallel BC$, so $ABCD$ is a parallelogram.

✓ Equal and parallel must be the **same pair**: $AB = DC$ and $AB \parallel DC$. (A trapezium can have $AB = DC$, $AD \parallel BC$.)

✗ $KLMN$ is not yet known to be a ∥gram, but "$\angle K = \angle M$ (opp. ∠s of ∥gram)".

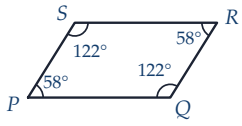
✓ You cannot use a property of the figure you are trying to prove. Find the angles another way (e.g. ∠ sum of polygon).

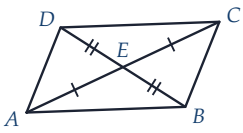
點解錯？

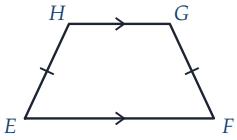
第一個：一對等長、另一對平行，可以係**等腰梯形**，唔一定係∥gram。
第二個：仲未證到係∥gram，就唔准用∥gram 嘅性質，否則係循環論證。

Quick quiz 快速測驗

Must each quadrilateral be a parallelogram? (answers at the foot of the page)







1 $PQRS$ **2** AEC, BED straight lines **3** $EF \parallel HG$, $EH = FG$

點樣做？

逐個條件對：①對邊等？②對角等？③對角線互相平分？④一對邊又等又平行？一個中都得；全部唔中就未必係。

Example 5 Level 1 $ABCD$ is a parallelogram. M and N are the mid-points of AB and DC. Prove that $MBND$ is a parallelogram.

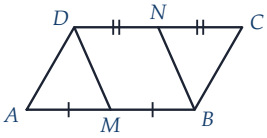
$MB = \frac{1}{2}AB$ and $DN = \frac{1}{2}DC$ (*given*)

$AB = DC$ (*opp. sides of ∥gram*)

$\therefore MB = DN$

Also $MB \parallel DN$ (*by definition*)

$\therefore$ MBND is a parallelogram (*2 sides equal and ∥*)



思路

MB 同 DN 喺兩條平行線上，**一定平行**；再證埋**等長**，就用條件 (d)。

Example 6 Level 1 Refer to the figure. (a) Find a. (b) Prove that $KLMN$ is a parallelogram.

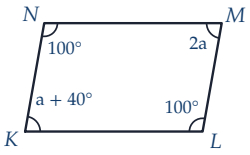
(a) $(a + 40°) + 100° + 2a + 100° = 360°$ (*∠ sum of polygon*)

$3a + 240° = 360°$, **$a = 40°$**

(b) $\angle K = 40° + 40° = 80°$, $\angle M = 2(40°) = 80°$ (*from (a)*)

$\angle K = \angle M$ and $\angle L = \angle N = 100°$

$\therefore$ KLMN is a parallelogram (*opp. ∠s equal*)



小心

(a) 只可以用「四邊形內角和 360°」。**唔可以**用 int. ∠s 或者 opp. ∠s of ∥gram，因為仲未證到佢係∥gram。

Answers (答案) **1** Yes (opp. ∠s equal) · **2** Yes (diags. bisect each other) · **3** No: an isosceles trapezium

Example 7 Level 2 $ABCD$ is a parallelogram. E and F are points on BD such that $BE = DF$. Prove that $AECF$ is a parallelogram.

Join AC, meeting BD at O.

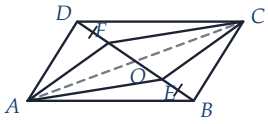
$OA = OC$ and $OB = OD$ (*diagonals of ∥gram*)

$OE = OB - BE$, $OF = OD - DF$

$\because OB = OD$ and $BE = DF$, $\therefore OE = OF$ (*given*)

AC and EF bisect each other at O.

$\therefore$ AECF is a parallelogram (*diags. bisect each other*)



思路

題目冇畀 AECF 嘅邊，但畀咗 BD 上面嘅點。「加一條線」：連 AC。AC 同 EF 係 AECF 嘅**對角線**，證佢哋互相平分就得。

另一個方法

用 SAS 證 △ABE ≅ △CDF 同 △AFD ≅ △CEB，得出兩對對邊相等 (opp. sides equal)。步驟長啲，但都啱。

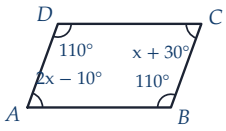
Which condition should I use? 揀邊個條件？

If the question gives you ...	try the condition ...
angles only	opp. ∠s equal
lengths on two pairs of opposite sides	opp. sides equal
points on the diagonals / mid-points of diagonals	diags. bisect each other
one pair of sides on two parallel lines	2 sides equal and ∥

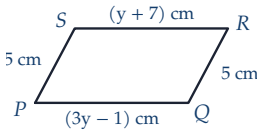
口訣

「有角諗角，有邊諗邊，有對角線就諗互相平分；邊喺平行線上，就諗又等又平行。」

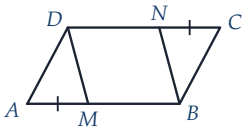
G Basic drilling 基本練習



1 (a) Find x. (b) Prove that $ABCD$ is a parallelogram.



2 Find y such that $PQRS$ is a parallelogram, and state the reason.



3 ∥gram $ABCD$, $AM = CN$. Prove that $MBND$ is a parallelogram.

Answers (答案) **1** (a) $2x - 10° + x + 30° + 220° = 360°$, $x = 40°$; (b) $\angle A = \angle C = 70°$, $\angle B = \angle D = 110°$ (opp. ∠s equal) · **2** $3y - 1 = y + 7$, $y = 4$; PQ = SR = 11 cm, PS = QR (opp. sides equal) · **3** MB = AB − AM = DC − CN = ND, MB ∥ ND (by definition), so MBND is a ∥gram (2 sides equal and ∥)

8.3 Rhombuses, Rectangles, Squares and Trapeziums

菱形、長方形、正方形及梯形

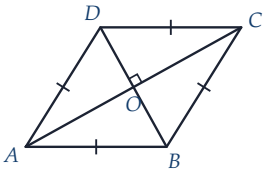
A rhombus, a rectangle and a square are all parallelograms, so they inherit every property you have just learnt. Each of them also adds something special of its own.

A Properties of rhombuses 菱形的性質

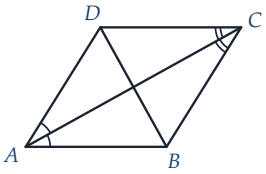
Concept 概念

A **rhombus**（菱形） is a quadrilateral with four equal sides. It has:

1. **all the properties of a parallelogram** (opposite sides, opposite angles, diagonals bisect each other);
2. diagonals that are **perpendicular**（互相垂直） to each other;
3. diagonals that **bisect the opposite angles**（平分對角）.



Diagonals ⊥: $\angle AOD = 90°$



Each diagonal bisects two angles

即係話

菱形係「加強版」平行四邊形：‖gram 有嘅佢全部有，仲多兩樣：對角線**成十字（90°）**，同埋對角線將每隻角**切開一半**。

生活化理解

風箏嘅骨架：兩支竹十字交叉，一定係直角。對摺菱形，兩邊會完全重疊，所以對角線將角平分。

理由寫法

兩個新性質都寫 **(property of rhombus)**。

Example 8 Level 1 *ABCD* is a rhombus. *AC* and *BD* meet at *O*. $\angle ADB = 70°$. Find *m*, *x* (= $\angle AOD$), *y* and *z*.

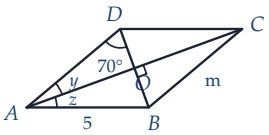
BC = *AB*, so **m = 5** (*by definition*)

$x = \angle AOD =$ **90°** (*property of rhombus*)

In $\triangle AOD$: $y + 90° + 70° = 180°$ (*∠ sum of △*)

y = 20°

$z = y =$ **20°** (*property of rhombus (AC bisects ∠DAB)*)



思路

見到菱形 + 對角線：即刻標 90°，再諗「一隻角切一半」。

Example 9 Level 2 A kite *ABCD* is a rhombus. Its diagonals meet at *E*, *AE* = 24 cm and *BD* = 20 cm. Find (a) the perimeter and (b) the area of the kite.

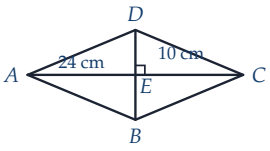
(a) *BE* = ½ × 20 = 10 cm, $\angle AEB = 90°$ (*property of rhombus*)

$AB = \sqrt{24^2 + 10^2} = 26$ cm (*Pyth. theorem*)

Perimeter = 4 × 26 = **104 cm** (*by definition*)

(b) Area = 2 × area of $\triangle ABD$ = 2 × ½ × 20 × 24

= **480 cm²**



小技巧

菱形面積 = ½ × 兩條對角線相乘。

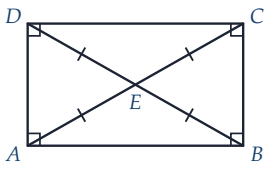
呢度：½ × 20 × 48 = 480 cm²，同答案一樣。

B Properties of rectangles 長方形的性質

Concept 概念

A **rectangle**（長方形） is a quadrilateral with four equal interior angles. It has:

1. **all the properties of a parallelogram**;
2. all interior angles are **right angles** (each $360° \div 4 = 90°$);
3. diagonals that are **equal** in length;
4. diagonals that cut each other into **four equal segments**: $AE = EB = EC = ED$.



即係話

長方形對角線**等長**又互相平分，所以由中心 E 去四個角一**樣遠**。

生活化理解

好似手機個 mon：中心點去四隻角距離一樣，所以 △ABE、△BCE ... 全部都係**等腰三角形**。

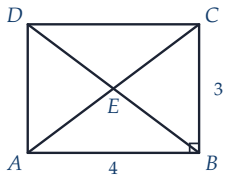
Example 10 Level 1 $ABCD$ is a rectangle. AC and BD meet at E. $AB = 4$ and $BC = 3$. Find EB.

$\angle ABC = 90°$ (*property of rectangle*)

$AC = \sqrt{4^2 + 3^2} = 5$ (*Pyth. theorem*)

$BD = AC = 5$ (*property of rectangle*)

$EB = \frac{1}{2}BD =$ **5/2** (*property of rectangle*)



思路

EB 係半條對角線。先用畢氏定理計 AC，再用「對角線等長、互相平分」。

Example 11 Level 2 $ABCD$ is a rectangle and $BCED$ is a parallelogram. ADE, AOC and BOD are straight lines. $CE = 10$ cm and $\angle AOB = 126°$. Find (a) AC, (b) $\angle AEC$.

(a) $AC = BD$ (*property of rectangle*)

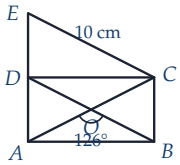
$BD = CE = 10$ cm, so **AC = 10 cm** (*opp. sides of ∥gram*)

(b) $OD = OA$, so $\angle ODA = \angle OAD$ (*property of rectangle; base ∠s, isos. △*)

$\angle ODA + \angle OAD = 126°$ (*ext. ∠ of △*)

$\angle ODA = 63°$

$\angle AEC = \angle ODA =$ **63°** (*corr. ∠s, CE ∥ BD*)



思路

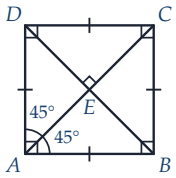
BD 有兩重身份：佢係長方形嘅**對角線**，亦係∥gram BCED 嘅一**條邊**。搵到呢條「橋」，(a) 就解決。

C Properties of squares 正方形的性質

Concept 概念

A **square**（正方形） has four equal sides **and** four equal angles, so it is both a rhombus and a rectangle. It has:

1. **all the properties of a rectangle** (right angles, equal diagonals);
2. **all the properties of a rhombus** (diagonals ⊥, diagonals bisect the angles);
3. diagonals that make **45°** with the sides.



即係話

正方形係「全能」：長方形、菱形嘅性質佢全部有。90° 嘅角俾對角線平分，就係 45°。

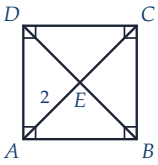
Example 12 Level 1 $ABCD$ is a square with diagonals meeting at E, and $AC = 2$. Find (a) BE, (b) the perimeter of $ABCD$ (leave $\sqrt{}$ in the answer).

(a) $AE = \frac{1}{2}AC = 1$, $BE = AE =$ **1** *(property of square)*

(b) $\angle AEB = 90°$ *(property of square)*

$AB = \sqrt{1^2 + 1^2} = \sqrt{2}$ *(Pyth. theorem)*

Perimeter $= 4 \times \sqrt{2} =$ **$4\sqrt{2}$** *(by definition)*



小技巧

正方形對角線切出嚟嘅四個三角形，全部係**等腰直角三角形**（45°–45°–90°）。

Example 13 Level 2 $ABCD$ is a square. $\angle CBE = 68°$ and $CE = DA$. Find x, y and z.

$CE = DA = CB$ *(given; by definition)*

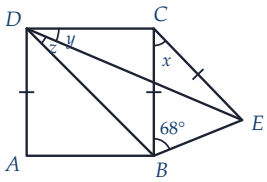
$\angle CEB = \angle CBE = 68°$ *(base ∠s, isos. △)*

$x = 180° - 68° - 68° =$ **44°** *(∠ sum of △)*

$\angle DCE = 90° + 44° = 134°$ *(property of square)*

$CD = CE$, so $2y + 134° = 180°$, **y = 23°** *(base ∠s, isos. △; ∠ sum of △)*

$\angle CDB = 45°$, so $z = 45° - 23° =$ **22°** *(property of square)*



思路

$CE = DA$，而正方形四邊相等，所以 **CE = CB = CD**。一個圖入面有**兩個等腰三角形**：△CBE 同 △CDE。

Quick check 即時練習

1 Square $PQRS$ has perimeter 4 cm. Its diagonals meet at V. Find PV.

2 Square $ABCD$: BC is produced to E with $CE = AC$. Find $\angle CAE$.

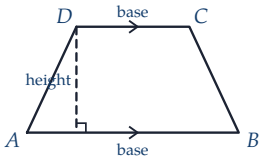
3 Rectangle $ABCD$: diagonals meet at O, $\angle AOB = 50°$. Find $\angle OAB$.

Answers（答案） **1** $PQ = 1$ cm, $PR = \sqrt{2}$ cm, $PV = \sqrt{2}/2$ cm · **2** $\angle ACE = 135°$, $\angle CAE = (180° - 135°)/2 = 22.5°$ · **3** OA = OB, $\angle OAB = (180° - 50°)/2 = 65°$

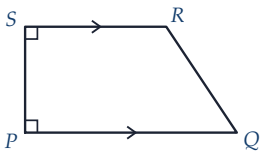
D Trapeziums 梯形

Concept 概念

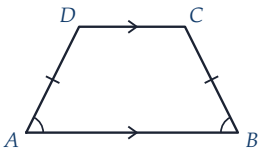
A **trapezium**（梯形） has only one pair of parallel sides. The parallel sides are the **bases**（底）; the perpendicular distance between them is the **height**（高）.



Bases and height



Right-angled trapezium
直角梯形 · a side ⊥ the bases



Isosceles trapezium
等腰梯形 · non-parallel sides equal

In an isosceles trapezium $ABCD$ ($AD \parallel BC$, $AB = DC$): **base angles**（底角） are equal ($\angle B = \angle C$, $\angle A = \angle D$) and the **diagonals are equal** ($AC = BD$).

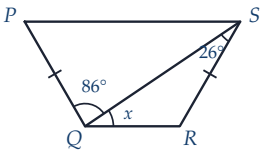
即係話

等腰梯形「左右對稱」：沿中間一條直線對摺，左右完全重疊。所以底角一樣、對角線一樣長。

小心

等腰梯形對角線**等長**，但**唔係**互相平分。同長方形唔同，唔好撈亂。

Example 14 Level 2 $PQRS$ is a trapezium, $PS \parallel QR$ and $PQ = SR$. $\angle PQS = 86°$ and $\angle QSR = 26°$. Find $\angle SQR$.



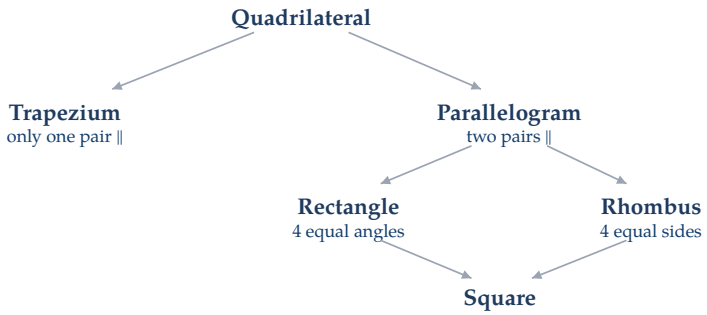
Let $\angle SQR = x$. $PQRS$ is an isosceles trapezium. (*given* $PQ = SR$)

$\angle SRQ = \angle PQR = x + 86°$ (*base ∠s of isos. trapezium*)

$x + 26° + (x + 86°) = 180°$ (*∠ sum of △*)

$2x = 68°$, $\boxed{\angle SQR = 34°}$

The family tree 四邊形家族圖



點睇？

箭咀向下 = 「下面嗰個有晒上面嗰個嘅所有性質」。

正方形喺最底：佢係長方形、菱形、平行四邊形。

梯形自己一支，**唔係**平行四邊形。

E Traps 陰濕位

✗ The diagonals of a rhombus are equal.	✓ They are **perpendicular** and bisect each other. Equal diagonals belong to a **rectangle**.
✗ The diagonals of a rectangle are perpendicular.	✓ They are **equal** and bisect each other. Perpendicular diagonals belong to a **rhombus**.
✗ A quadrilateral with four equal angles must be a square.	✓ Four equal angles only gives a **rectangle**. A square also needs four equal sides.
✗ Using "base ∠s of isos. trapezium" before showing $PQ = SR$.	✓ First state the equal non-parallel sides (given), then use the property.

點分？

菱形：對角線「十字」(⊥)。

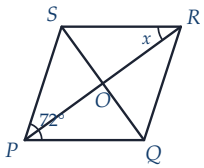
長方形：對角線「一樣長」。

正方形：兩樣都有。

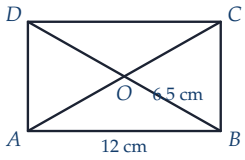
口訣

「菱形講垂直，長方講等長，正方樣樣齊。」

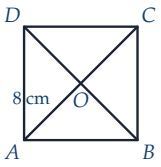
F Basic drilling 基本練習



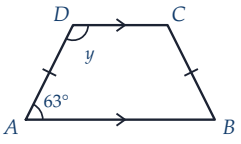
1 Rhombus $PQRS$, $\angle SPQ = 72°$. Find x.



2 Rectangle $ABCD$, $OB = 6.5$ cm, $AB = 12$ cm. Find BC.



3 Square $ABCD$, $AC = 8$ cm. Find OB and $\angle OBC$.



4 Isosceles trapezium, $AB \parallel DC$, $\angle A = 63°$. Find $\angle B$ and $y = \angle D$.

Answers（答案） **1** $\angle QRS = 72°$ (opp. ∠s of ∥gram), $x = 36°$ (property of rhombus) · **2** $AC = BD = 13$ cm, $BC = \sqrt{13^2 - 12^2} = 5$ cm · **3** $OB = 4$ cm, $\angle OBC = 45°$ · **4** $\angle B = 63°$ (base ∠s), $y = 117°$ (int. ∠s, AB ∥ DC)

8.4 Proofs Related to Parallelograms

平行四邊形的相關證明

Now we put everything together. A geometric proof is a chain: each line follows from a given fact, a property, or a line already proved. Learning to build the chain is the goal of this section.

A How to write a proof 點樣寫證明

A four-step routine 四步曲

1. **Mark the figure.** Copy every given fact onto the figure: ticks, arcs, ∥ arrows, right angles.
2. **Work backwards.** Ask: "To prove $AF = BE$, which two triangles contain them?"
3. **Choose the tool.** Congruent triangles (SSS, SAS, ASA, AAS, RHS), or a condition for a parallelogram.
4. **Write one fact per line**, each with its reason. End with the conclusion and its reason.

即係話

證明好似砌積木：每一粒都要有根據（given、性質、或者上面證過嘅）。唔可以「睇落係」就當係。

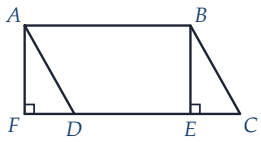
口訣

「標圖、倒推、揀工具、逐行寫理由。」

Example 15 Level 1 $ABCD$ is a parallelogram and $FDEC$ is a straight line. $AF \perp FC$ and $BE \perp FC$. Prove that $AF = BE$.

In $\triangle AFD$ and $\triangle BEC$:

$\angle AFD = \angle BEC = 90°$	(*given*)
$\angle ADF = \angle BCE$	(*corr. $\angle$s, $AD \parallel BC$*)
$AD = BC$	(*opp. sides of ∥gram*)
$\therefore \triangle AFD \cong \triangle BEC$	(*AAS*)
$\therefore AF = BE$	(*corr. sides, $\cong\triangle$s*)



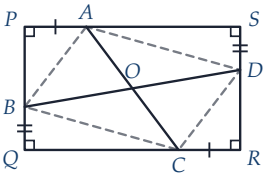
思路

AF 同 BE 分別喺 △AFD 同 △BEC 入面。兩隻直角 + 一對同位角 + 一對邊 → AAS。

Example 16 Level 2 $PQRS$ is a rectangle. AC and BD meet at O. $AP = CR$ and $BQ = DS$. Prove that AC and BD bisect each other.

Join AB, BC, CD, DA.

$PB = PQ - BQ = SR - DS = RD$	(*property of rectangle; given*)
$\triangle APB \cong \triangle CRD$ ($AP = CR$, $\angle P = \angle R = 90°$, $PB = RD$)	(*SAS*)
$\therefore AB = CD$	(*corr. sides, $\cong\triangle$s*)
Similarly $\triangle BQC \cong \triangle DSA$, $\therefore BC = DA$	(*SAS; corr. sides, $\cong\triangle$s*)
$\therefore ABCD$ is a parallelogram	(*opp. sides equal*)
$\therefore$ AC and BD bisect each other	(*diagonals of ∥gram*)



思路

要證「互相平分」，最順手係先證 ABCD 係∥gram，再用「∥gram 對角線互相平分」。**判定**同**性質**一齊用。

Example 17 Level 2 $ABCD$ is a square. $AOCE$ and BOD are straight lines. (a) Prove that $BE = DE$. (b) If $DE = DB$, find $\angle EBC$.

(a) In $\triangle BOE$ and $\triangle DOE$:

$BO = DO$ (*property of square*)

$\angle BOE = \angle DOE = 90°$ (*property of square*)

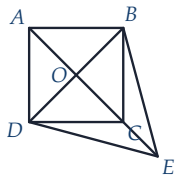
$OE = OE$ (*common side*)

$\therefore \triangle BOE \cong \triangle DOE$, $\boxed{BE = DE}$ (*SAS; corr. sides, $\cong\triangle$s*)

(b) $DE = DB = BE$, so $\triangle BDE$ is equilateral (*given; proved in (a)*)

$\angle DBE = 60°$, $\angle DBC = 45°$ (*property of equil. $\triangle$; property of square*)

$\angle EBC = 60° - 45° = \boxed{15°}$



思路

(b) 用返 (a) 嘅結果：**上一部分證到嘅嘢，下一部分可以用**，理由寫 (proved in (a))。

Traps 陰濕位

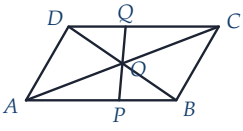
✗ $\triangle AFD \cong \triangle BEC$ (AAS), with no lines listing the three pairs.	✓ List the three pairs of equal parts, **each with a reason**, before writing "$\cong$".
✗ Letters out of order: $\triangle AFD \cong \triangle CEB$.	✓ Match the vertices in order: $A\leftrightarrow B$, $F\leftrightarrow E$, $D\leftrightarrow C$, so $\triangle AFD \cong \triangle BEC$.
✗ Using AAA or SSA as a congruence reason.	✓ Only SSS, SAS, ASA, AAS and RHS prove congruence.

點解錯？

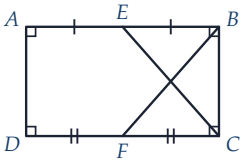
全等符號 $\cong$ 嘅字母次序要**對應**，A 對 B、F 對 E...，次序錯咗之後「corr. sides」都會錯。

AAA 只係相似，唔係全等；SSA 唔一定成立。

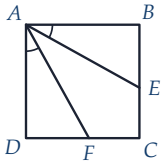
B Basic drilling 基本練習



1 ∥gram $ABCD$. AOC, BOD, POQ are straight lines. Prove that $PO = QO$.



2 Rectangle $ABCD$, $AE = EB$, $DF = FC$. Prove that $BF = CE$.



3 Square $ABCD$, $\angle BAE = \angle DAF$. Prove that $\triangle AEF$ is isosceles.

Answers （答案） **1** OA = OC (diagonals of ∥gram), ∠PAO = ∠QCO (alt. ∠s, AB ∥ DC), ∠AOP = ∠COQ (vert. opp. ∠s); △AOP ≅ △COQ (ASA), PO = QO · **2** EB = FC (half of equal sides), ∠EBC = ∠FCB = 90°, BC common; △EBC ≅ △FCB (SAS), CE = BF · **3** AB = AD, ∠ABE = ∠ADF = 90°, ∠BAE = ∠DAF; △ABE ≅ △ADF (ASA), AE = AF

8.5 Mid-point Theorem and Intercept Theorem

中點定理及截線定理

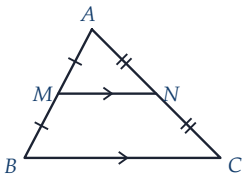
Using what we know about parallelograms, we can now prove two theorems about triangles and parallel lines. They let us find lengths and prove lines parallel without measuring anything.

A Mid-point theorem 中點定理

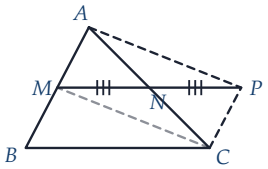
Theorem 定理

The straight line joining the **mid-points**（中點） of two sides of a triangle is **parallel** to the third side and equal to **half** of it.

In $\triangle ABC$, if $AM = MB$ and $AN = NC$, then (a) $MN \parallel BC$, (b) $MN = \frac{1}{2}BC$. [Reference: (mid-pt. theorem)]



Given: two mid-points
Result: $MN \parallel BC$, $MN = \frac{1}{2}BC$



Proof idea: produce MN to P with $MN = NP$

即係話

三角形兩條邊各自搵中點，連埋佢哋：呢條線**同第三邊平行**，而且**長度係佢一半**。

生活化理解

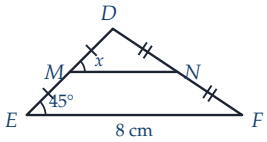
好似將三角形「攔腰切一刀」：上面細三角形同原本個形狀一樣，但係邊長全部縮細一半。

點證？

延長 MN 到 P，令 MN = NP。AMCP 對角線互相平分 → ∥gram；再得 MBCP 係 ∥gram（2 sides equal and ∥），所以 MN ∥ BC 同 BC = MP = 2MN。

Example 18 Level 1 M and N are the mid-points of DE and DF. $EF = 8$ cm and $\angle DEF = 45°$. Find x and MN.

$DM = ME$ and $DN = NF$	(*given*)
$\therefore MN \parallel EF$	(*mid-pt. theorem*)
$x = \angle DEF =$ **45°**	(*corr. ∠s, MN ∥ EF*)
$MN = \frac{1}{2}EF = \frac{1}{2} \times 8$ cm = **4 cm**	(*mid-pt. theorem*)



Key points 重點記住

1. You need **two** mid-points on **two** sides of the **same** triangle.
2. The theorem gives **two** results: the line is **parallel**, and it is **half** the third side.

口訣

「兩個中點一條線，平行第三邊，長度得一半。」

Example 19 LEVEL 2 D is the mid-point of both AC and BF. E is the mid-point of AB. GCB is a straight line, $GC = 4$ cm, $DE = 2$ cm and $\angle FGB = 65°$. Find (a) CB, (b) $\angle ADE$.

(a) In $\triangle ABC$: $AD = DC$, $AE = EB$ *(given)*

$DE = \frac{1}{2}CB$, so $CB = 2 \times 2 =$ **4 cm** *(mid-pt. theorem)*

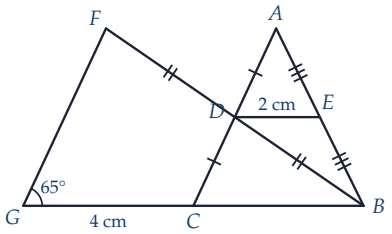
(b) In $\triangle BFG$: $BD = DF$, $BC = CG = 4$ cm *(given; proved in (a))*

$\therefore DC \parallel FG$ *(mid-pt. theorem)*

$\angle ACB = \angle FGB = 65°$ *(corr. $\angle$s, $DC \parallel FG$)*

In $\triangle ABC$: $DE \parallel CB$ *(mid-pt. theorem)*

$\angle ADE = \angle ACB =$ **65°** *(corr. $\angle$s, $DE \parallel CB$)*



思路

一個圖有**兩個**三角形可以用中點定理：△ABC 同 △BFG。(a) 計到 CB = 4 = GC，所以 C 都係中點，(b) 就再用多次。

Example 20 LEVEL 2 E, F, G and H are the mid-points of AB, BC, CD and DA. Prove that $EFGH$ is a parallelogram.

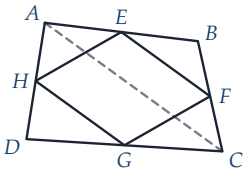
Join AC.

In $\triangle ABC$: $EF \parallel AC$ and $EF = \frac{1}{2}AC$ *(mid-pt. theorem)*

In $\triangle ADC$: $HG \parallel AC$ and $HG = \frac{1}{2}AC$ *(mid-pt. theorem)*

$\therefore EF \parallel HG$ and $EF = HG$

$\therefore$ EFGH is a parallelogram *(2 sides equal and $\parallel$)*



有趣結果

任何四邊形，連埋四條邊嘅中點，都一定得到平行四邊形。試吓自己畫一個歪歪地嘅四邊形驗證。

QUICK CHECK 即時練習

1 In $\triangle ABC$, D and E are the mid-points of AB and AC. $BC = 14$ cm. Find DE.

2 In $\triangle PQR$, X, Y are mid-points of PQ, PR and $\angle PQR = 52°$. Find $\angle PXY$.

3 $ABCD$ is a rectangle. E, F, G, H are mid-points of its sides. Prove $EFGH$ is a rhombus (idea only).

ANSWERS （答案） **1** $DE = 7$ cm (mid-pt. theorem) · **2** $XY \parallel QR$, $\angle PXY = 52°$ (corr. $\angle$s, $XY \parallel QR$) · **3** Each side of EFGH = ½ × a diagonal (mid-pt. theorem); AC = BD, so all four sides equal

B Intercept theorem 截線定理

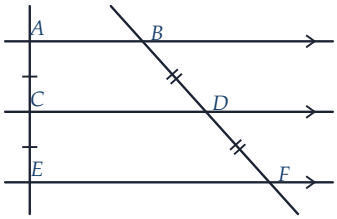
Terms 名詞

A **transversal**（截線） is a line crossing several lines. The length of the segment cut off between two of the lines is an **intercept**（截距）.

Theorem 定理

If three or more **parallel** lines make **equal intercepts** on one transversal, they make equal intercepts on **any other** transversal.

If $AB \parallel CD \parallel EF$ and $AC = CE$, then $BD = DF$. [Reference: (intercept theorem)]



即係話

三條平行線，喺第一條斜線上切出**一樣長**嘅兩段，咁喺第二條斜線上切出嚟嘅兩段**都會一樣長**。

生活化理解

原子筆紙上面嘅橫間線係平行同等距。用間尺斜斜放上去，每一格喺間尺上量到嘅長度都一樣。

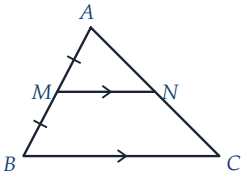
小心

兩條截線上嘅截距**唔一定**等於對方：AC = CE，BD = DF，但係 AC 唔一定等於 BD。

Special case in a triangle 三角形特例

If a line through the **mid-point** of one side of a triangle is **parallel** to another side, it **bisects** the third side.

In $\triangle ABC$, if $AM = MB$ and $MN \parallel BC$, then $AN = NC$. (intercept theorem)



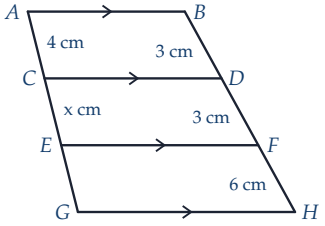
Example 21 Level 1 $ACEG$ and $BDFH$ are straight lines with $AB \parallel CD \parallel EF \parallel GH$. Find x and EG.

$BD = DF = 3$ cm (*given*)

$\therefore CE = AC$, $\boxed{x = 4}$ (*intercept theorem*)

$BF = 3 + 3 = 6$ cm $= FH$

$\therefore EG = AE = 4 + 4 = \boxed{8 \text{ cm}}$ (*intercept theorem*)



思路

第二步用嘅係 AB、EF、GH 三條平行線，**截距**係 BF 同 FH。揀啱邊三條線好重要。

Example 22 LEVEL 2 AED and BFC are straight lines, $AB \parallel EF \parallel DC$, $AE = ED$, $AB = 26$ cm and $EF = 34$ cm. BD meets EF at T. (a) Prove that $BT = TD$. (b) Is T the mid-point of EF?

(a) In $\triangle DAB$: $AE = ED$ and $ET \parallel AB$ *(given)*

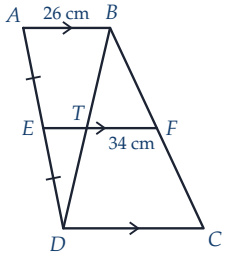
$\therefore$ BT = TD *(intercept theorem)*

(b) In $\triangle DAB$: $AE = ED$, $BT = TD$ *(given; proved in (a))*

$ET = \frac{1}{2}AB = 13$ cm *(mid-pt. theorem)*

$\frac{1}{2}EF = 17$ cm $\neq 13$ cm

T is not the mid-point of EF



思路

(a) 得**一個**中點 + **平行** → 截線定理。(b) 而家有**兩個**中點 → 中點定理計 ET。

Example 23 LEVEL 2 F is the mid-point of BC. D and E divide AB into three equal parts. AF and DC meet at G. (a) Prove that $AG = GF$. (b) Find $DG : DC$.

(a) In $\triangle BCD$: $BE = ED$, $BF = FC$, $\therefore EF \parallel DC$ *(given; mid-pt. theorem)*

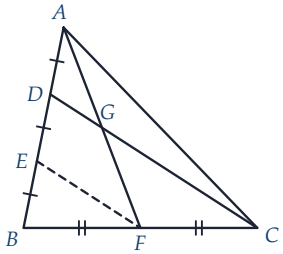
In $\triangle AEF$: $AD = DE$ and $DG \parallel EF$, *(given; intercept theorem)*

$\therefore$ AG = GF

(b) $EF = \frac{1}{2}DC$ *(mid-pt. theorem ($\triangle BCD$))*

$DG = \frac{1}{2}EF = \frac{1}{4}DC$ *(mid-pt. theorem ($\triangle AEF$))*

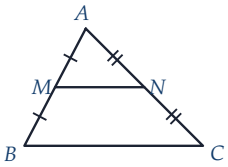
DG : DC = 1 : 4



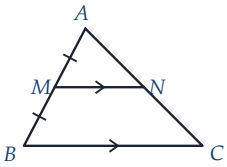
思路

加一條**輔助線** EF。△BCD 同 △AEF 輪流用兩個定理。

DO NOT MIX THEM UP 兩個定理點分？



Mid-point theorem
Know: **two mid-points**
Get: $MN \parallel BC$ **and** $MN = \frac{1}{2}BC$



Intercept theorem (triangle)
Know: **one mid-point + parallel**
Get: $AN = NC$

口訣

「兩個中點 → 中點定理；一個中點加平行 → 截線定理。」

小心

用中點定理之前，要確認兩個點真係**中點**（有 given 或者已經證到）。

C Traps 陰濕位

✗ $AM = MB$ and $MN \parallel BC$, $\therefore MN = \frac{1}{2}BC$ (mid-pt. theorem).

✓ One mid-point + $\parallel$ gives $AN = NC$ (intercept theorem) first. Only then use the mid-point theorem for $MN = \frac{1}{2}BC$.

✗ Intercepts on two transversals are equal to each other: $AC = BD$.

✓ Equal intercepts on **one** transversal give equal intercepts on the **other**: $AC = CE \Rightarrow BD = DF$.

✗ Using the intercept theorem when the lines are not stated parallel.

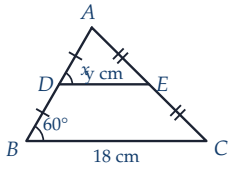
✓ The intercept theorem needs **three or more parallel lines**. Check for $\parallel$ first.

點解錯？

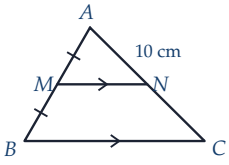
第一個：得一個中點，未知 N 係咪中點，**唔可以**直接用中點定理。要先用截線定理證 N 係中點。

第二個：截距只係同一條截線上面相比。

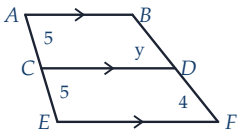
D Basic drilling 基本練習



1 D, E are mid-points of AB, AC. Find x and y.



2 $AM = MB$, $MN \parallel BC$, $AC = 10$ cm. Find NC.



3 $AB \parallel CD \parallel EF$. Find y.

4 In $\triangle PQR$, X, Y are mid-points of PQ, PR. $XY = 3.5$ cm. Find QR.

5 The mid-points of the sides of a rectangle are joined. What shape is formed? Why?

6 $\triangle ABC$ has perimeter 24 cm. Find the perimeter of the triangle formed by joining the mid-points of its sides.

Answers （答案） **1** $DE \parallel BC$, $x = 60°$ (corr. ∠s, DE ∥ BC); $y = 9$ (mid-pt. theorem) · **2** $AN = NC$ (intercept theorem), $NC = 5$ cm · **3** $AC = CE = 5$, so $BD = DF$ (intercept theorem), $y = 4$ · **4** $QR = 2 \times 3.5 = 7$ cm (mid-pt. theorem) · **5** A rhombus: each side = ½ × a diagonal of the rectangle, and the diagonals are equal · **6** 12 cm (each side is half of a side of △ABC)

Summary Chapter 8 at a Glance

本章總結

One page to revise everything: which properties each quadrilateral has, how to prove a parallelogram, and the two theorems about mid-points and parallel lines.

A Properties of special quadrilaterals 特殊四邊形性質一覽

Property	∥gram	Rhombus	Rectangle	Square	Isos. trap.
Two pairs of opposite sides ∥	✓	✓	✓	✓	—
Opposite sides equal	✓	✓	✓	✓	—
All four sides equal	—	✓	—	✓	—
Opposite angles equal	✓	✓	✓	✓	—
All angles are 90°	—	—	✓	✓	—
Diagonals bisect each other	✓	✓	✓	✓	—
Diagonals are equal	—	—	✓	✓	✓
Diagonals are perpendicular	—	✓	—	✓	—
Diagonals bisect the angles	—	✓	—	✓	—
Base angles equal	—	—	—	—	✓

B Conditions for a parallelogram 判定平行四邊形

Show that ...	Reason
both pairs of opposite sides are ∥	(by definition)
both pairs of opposite sides are equal	(opp. sides equal)
both pairs of opposite angles are equal	(opp. ∠s equal)
the diagonals bisect each other	(diags. bisect each other)
one pair of sides is equal **and** ∥	(2 sides equal and ∥)

口訣總匯

「F 同 Z 一樣大，U 字加埋一百八。」

「對邊等、對角等、對角線互相平分。」

「菱形講垂直，長方講等長，正方樣樣齊。」

「兩個中點 → 中點定理；一個中點加平行 → 截線定理。」

C Two theorems 兩個定理

Theorem	If ...	then ...
Mid-point theorem	$AM = MB$ and $AN = NC$	$MN \parallel BC$, $MN = \frac{1}{2}BC$
Intercept theorem	$AB \parallel CD \parallel EF$ and $AC = CE$	$BD = DF$
Intercept (triangle)	$AM = MB$ and $MN \parallel BC$	$AN = NC$

溫習建議

考試前用呢頁做「自我測驗」：遮住右邊，講出性質同理由寫法。